COMP396 Weekly Meeting Minutes

(Week 3)

Date & Time: Wednesday, February 11th, 2026, 11:00–13:00

Location: Harold Cohen Library, 3rd Floor, Group Study Room 3F

Note Taker: Zhao Saria (ZS)

Attendees: Zhang Olivia (ZO), Zhao Saria (ZS), Xue Ella (XE), Wong Haw (HW)

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1. Agenda

- Review the completion status of the Task A series (A1–A4).
- Formally finalize the selection of the three core independent strategies.
- Document the specific technical rationales for rejecting alternative strategies (MVO, HRP, etc.).
- Allocate tasks for the asset-to-strategy mapping phase.

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2. Discussion & Decisions

2.1. Task Review and Data Verification

Following a secondary analysis by HW, ZO, and XE, the team verified that the structural characteristics of the current ten asset series (Series 01–10) are fully consistent with the preliminary findings presented by ZS in Week 2.

2.2. Final Strategy Selection

After evaluating multiple quantitative models, the team has officially selected the following three strategies for the Part 2 implementation:

- **Trend/Momentum Strategy:** Targeting assets with strong directional growth.
- **Mean-Reversion Strategy:** Refined logic based on the Phase 1 model for range-bound assets.
- **Defensive/Short-selling Strategy:** Specifically designed to capitalize on assets with clear downward trajectories.

2.3. Rationale for Strategy Rejections

To ensure the academic rigor of the upcoming Design Document, the team detailed why the following models were not adopted:

I. Mean-Variance Optimization (MVO):

- **Input Sensitivity:** Mean-Variance Optimization is highly sensitive to expected returns and covariance matrices; with only 10 series, small estimation errors lead to extreme weight concentration.
- **Risk Metric Flaws:** MVO treats "variance" as the sole risk, failing to distinguish between "good volatility" (directional profit) and "bad volatility" (unstructured noise).
- **Logic Mismatch:** MVO optimizes weights globally rather than matching specific strategies to asset structures, which contradicts our "structure-matching" core logic.

II. Hierarchical Risk Parity (HRP):

- **Environment Mismatch:** HRP is designed to mitigate concentration risk in highly correlated asset pools; however, our 10 assets show distinct structural differences, making HRP's clustering redundant.
- **Research Focus:** Our project prioritizes exploiting structural differences through specific strategies rather than purely diversifying risk at the portfolio level.
- **Sample Size Constraints:** Given the limited sample length of the 10 assets, the clustering stability of HRP is insufficient, adding complexity without significant marginal gain.

III. Regime-Switching Conditional Probability Strategy:

- **Overfitting Risk:** The excessive structural complexity (switching rules, thresholds, etc.) creates a high risk of "out-of-sample" collapse in Part 3.

- **Weak Prediction:** Daily conditional probability often hovers near 0.5, providing a thin "edge" easily negated by slippage.
- **Stability:** Unlike probability strategies that flip directions frequently, Trend Consistency models using Moving Averages are more likely to achieve a high PD Ratio.

IV. Volatility-Filtered Momentum Strategy:

- This focuses on risk-adjustment rather than structural identification. Filtering by volatility may exclude high-volatility/high-return assets (e.g., Series 04) and introduces extra parameters, increasing the risk of overfitting on limited data.

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3. Task Allocation

Task ID	Task Description	Owner	Due Date
B1	Evaluate and select the optimal asset series for the **Trend/Momentum strategy**.	ZS	18.Feb
B2	Evaluate and select the optimal asset series for the **Mean-Reversion strategy**.	ZO	18. Feb
B3	Evaluate and select the optimal asset series for the **Defensive/Short-selling strategy**.	XE	18. Feb
B4	Evaluate the feasibility of the Phase 1 **CVaR** model for portfolio strategy and research 2-3 alternative combination strategies for future selection.	WH	20. Feb

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4. Current Challenges and Issues

- **Asset-Strategy Mapping Accuracy:** Although the team has identified the structural characteristics of the ten series (e.g., Series 05 as a downward trend), the final performance depends on selecting the precise assets that maximize the Single-Series PD Ratio. A challenge remains in determining whether a "noise-heavy" asset like Series 10 should be included or excluded to

protect the Aggregate PD Ratio.

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5. Next Meeting Arrangement

- **Date & Time:** Friday, February 20th, 2026, 14:00–16:00
- **Location:** Tencent Meeting (Online)
- **Key Agenda Items:**
    - Confirm the final asset-to-strategy mapping based on B1–B3 results.
    - Review HW's research on portfolio integration (CVaR vs. new alternatives).
    - Finalize the coding timeline for the T-Strategy class development.